Machine Learning Algorithms

K-MEANS CLUSTERING
KNN
Naive Baise

Pros and cons

Advantages of k-means

1. Relatively simple to implement.
2. Scales to large data sets.
3. Guarantees convergence.
4. Easily adapts to new examples.

Disadvantages of k-means

1. Choosing \(k\) manually.
2. Being dependent on initial values.
3. Scaling with number of dimensions.

Pros and Cons

Advantages of Naive Bayes

1. This algorithm works very fast and can easily predict the class of a test dataset.
2. You can use it to solve multi-class prediction problems as it's quite useful with them.
3. Naive Bayes classifier performs better than other models with less training data if the assumption of independence of features holds.

Disadvantages of Naive Bayes

1. If your test data set has a categorical variable of a category that wasn't present in the training data set, the Naive Bayes model will assign it zero probability and won't be able to make any predictions in this regard.
2. It assumes that all the features are independent. While it might sound great in theory, in real life, you'll hardly find a set of independent features.

KNN pros and cons

Pros

1. No Training Period: KNN is called Lazy Learner (Instance based learning).
2. new data can be added seamlessly which will not impact the accuracy of the algorithm.
3. KNN is very easy to implement, There are only two parameters required to implement KNN i.e. the value of K and the distance function (e.g. Euclidean or Manhattan etc.)

Cons

1. Does not work well with large dataset:
2. Does not work well with high dimensions
3. Need feature scaling: We need to do feature scaling (standardization and normalization)
4. Sensitive to noisy data, missing values.

Decision Tree pros and cros

Advantages:

1. Compared to other algorithms decision trees requires less effort for data preparation during pre-processing.
2. A decision tree does not require normalization of data.
3. A decision tree does not require scaling of data as well.
4. Missing values in the data also do NOT affect the process of building a decision tree to any considerable extent.
5. A Decision tree model is very intuitive and easy to explain to technical teams as well as stakeholders.

Disadvantage:

6. A small change in the data can cause a large change in the structure of the decision tree causing instability.
7. For a Decision tree sometimes calculation can go far more complex compared to other algorithms.
3. Decision tree often involves higher time to train the model.
4. Decision tree training is relatively expensive as the complexity and time has taken are more.
5. The Decision Tree algorithm is inadequate for applying regression and predicting continuous values.